

Targeted Small RNA-Seq Analysis Report

- Alignment and Quality Control
- Molecule Level Estimates
- Targeted Gene Enrichment
- Candidate Variant Analysis
- Soft-Clipped Read Characterization

Prepared for SOPHiA GENETICS as part of the
Senior NGS Bioinformatician Technical Assessment

Joshua Rivera, Ph.D.
Dana-Farber Cancer Institute / Harvard Medical School
May 2026

This report presents a stakeholder-facing technical assessment of targeted small RNA sequencing data provided as part of the SOPHiA GENETICS Senior NGS Bioinformatician evaluation process.

Outline

1. Abstract
2. Quality Control
3. Targeted Enrichment and Deduplication
4. Candidate Variant Analysis
5. Soft-Clipped Read Characterization
6. Discussion
7. Conclusion
8. Reproducibility and Supplemental Materials

ABSTRACT

This report presents an analysis of a targeted small RNA sequencing dataset consisting of three hg19-aligned BAM files. The dataset was generated using an Illumina NextSeq targeted sequencing workflow designed to enrich a restricted subset of genomic loci. Reads contained embedded 8-mer molecular barcodes stored within the mb BAM tag, enabling barcode deduplication and direct quantification of amplification redundancy. Coverage was strongly localized to a relatively small set of genomic regions, consistent with a targeted enrichment strategy rather than genome-wide sequencing.

The primary objectives of this analysis were to:

1. Evaluate sequencing and alignment quality of the BAM files,
2. Identify enriched genes and targeted genomic loci,
3. Characterize candidate genetic variants,
4. Investigate the origin and interpretation of large soft-clipped read segments.

Across all analyses, the dataset showed strong alignment quality and reproducible enrichment across a focused set of cancer-associated genes, including CCND1, AXL, MET, EGFR, FGFR1, BRAF, and HRAS. Barcode deduplication revealed substantial differences in molecular complexity between samples, with sample3 exhibiting elevated amplification redundancy and the largest reduction in signal following deduplication. Candidate variant analysis identified a small number of recurrent events, including two adjacent FGFR1 substitutions detected across all three samples. Soft-clipped read analysis further showed that these events were concentrated within a limited number of recurrent loci rather than being distributed uniformly throughout the genome.

Molecular barcode information proved critical for distinguishing amplification associated signal bias from true molecular abundance and provided additional context for interpreting variant and soft-clipping patterns.

RESULTS

Quality Control

Mapping quality reads

The primary objective of the quality control (QC) analysis is to evaluate the technical reliability, molecular complexity, and alignment behavior of the targeted small RNA-seq dataset before moving onto downstream interpretation. The data provided was already aligned BAM files; thus, the QC workflow focuses on alignment quality and molecule level characteristics.

Metrics for observation include, total sequencing depth, fraction of mapped and uniquely mapped reads, mapping quality (MAPQ) distributions, per sample read length distributions, and barcode complexity.

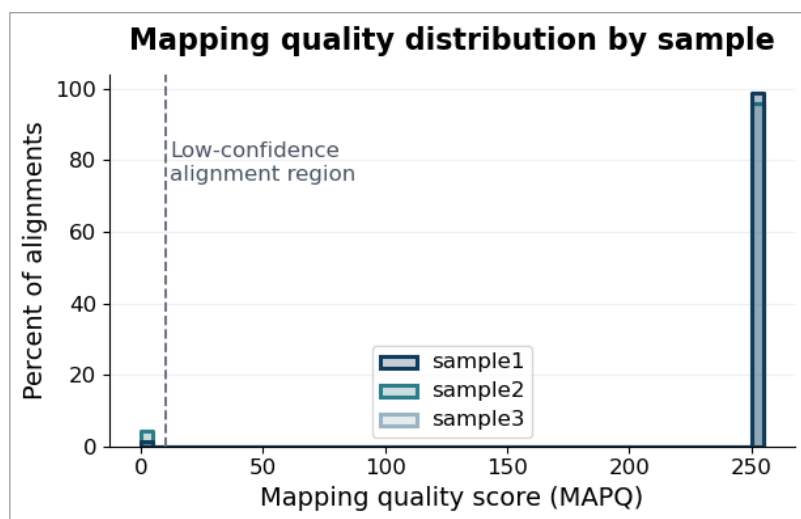


Figure 1. Mapping quality distribution across samples. Histogram showing the distribution of alignment mapping quality (MAPQ) scores across all three targeted small RNA sequencing samples.

Initial read-level BAM inspection confirmed that the dataset consisted primarily of single-end alignments containing embedded molecular barcode annotations (mb tags). Mapping quality analysis demonstrated that most alignments clustered at high-confidence values (MAPQ=255), consistent with predominantly unique or confidently resolved

alignments across enriched target regions (Figure 1).

Only a relatively small fraction of reads occupied the low-MAPQ region, indicating limited multimapping or ambiguous alignment behavior. These findings support a technically

coherent targeted sequencing dataset with generally strong alignment confidence across all three samples. Now we will inspect how many reads confidently mapped to the genome.

Alignment composition analysis

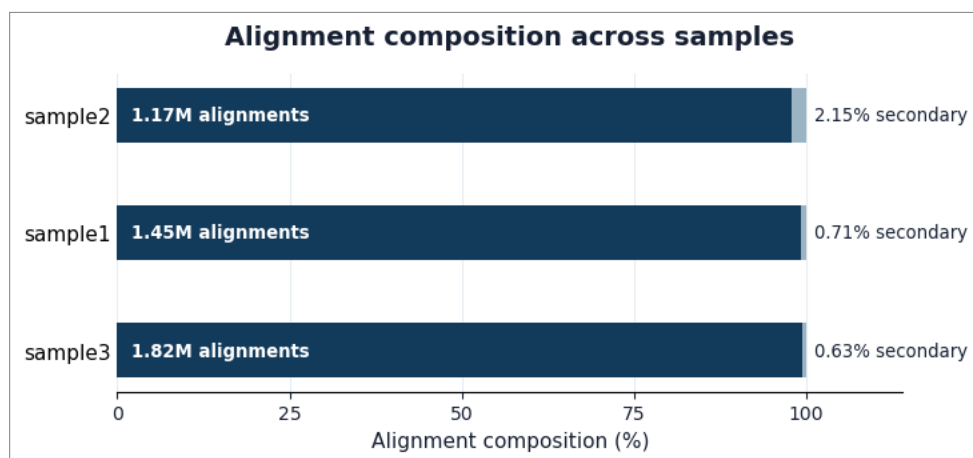


Figure 2. Alignment composition across samples. Horizontal proportional bars showing the fraction of primary and secondary alignments across all three targeted small RNA sequencing samples.

Alignment composition analysis demonstrated that primary alignments accounted for the overwhelming majority of reads across all three samples, supporting generally strong alignment specificity throughout the targeted sequencing dataset (Figure 2). Despite the

targeted enrichment design and localized coverage structure, secondary alignment fractions remained low overall, suggesting that most reads resolved unambiguously to enriched target regions.

Sample2 demonstrated a modestly elevated secondary alignment fraction relative to sample1 and sample3, potentially reflecting conserved domain homology or target or reduced alignment specificity within a subset of enriched loci. However, secondary alignments remained a minor component of the overall sequencing signal across all samples.

Read length distribution analysis

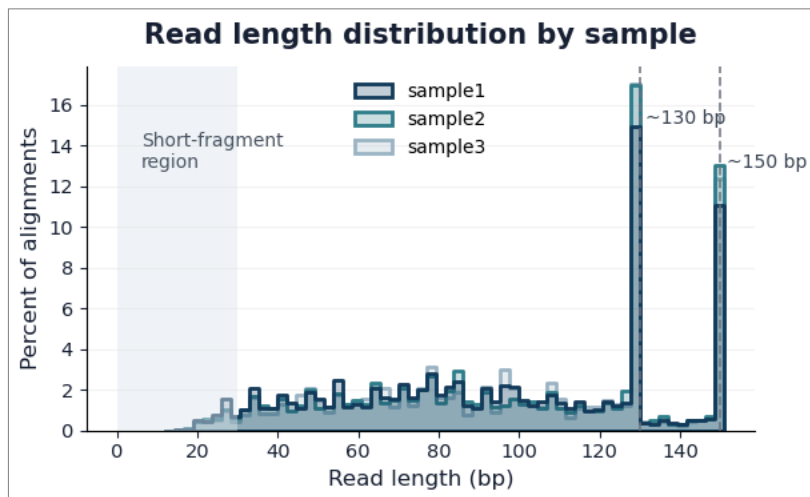


Figure 3. Read length distribution across samples. Step histograms showing reproducible read length structure across all three targeted small RNA sequencing samples. Prominent enrichment peaks centered near ~130 nt and ~150 nt

Read length distributions were highly reproducible across samples, with prominent enrichment peaks centered around ~130 nt and ~150 nt. The consistency of these peaks across all three samples suggests structured, non-random fragment populations associated with the targeted sequencing assay rather than broad stochastic fragmentation (Figure 3).

Because the analysis was performed on post-aligned BAM files without access to upstream adapter-trimming metadata or raw FASTQ preprocessing steps, these values will be interpreted as aligned sequencing fragment lengths rather than direct estimates of native RNA insert size alone. In addition to the dominant enrichment peaks, a smaller population of shorter fragments (~20–100 nt) was also observed, potentially representing partially processed, truncated, or structurally distinct sequencing products associated with enriched loci.

An additional feature of the read length distribution was the apparent depletion of fragments between the dominant ~130 nt and ~150 nt enrichment peaks. Rather than forming a single broad fragment population, the data suggest the presence of at least two distinct length classes separated by a relative underrepresentation of intermediate-sized fragments. Although the biological or technical origin of this pattern cannot be determined from the available post-aligned BAM files alone, the reproducibility of the feature across all three samples suggests it is unlikely to reflect random sampling variation.

Barcode Complexity

To evaluate molecular complexity, I leveraged the embedded 8-mer molecular barcode (mb) annotations contained within the BAM records. Because targeted sequencing depth can be inflated by repeated amplification of the same originating molecule, barcode deduplication was used to distinguish unique molecular observations from amplification-derived duplicates.

Amplification redundancy was quantified as the ratio of primary alignments to unique molecular barcodes within each sample. This metric estimates the number of sequencing observations generated from each unique molecular barcode, with larger values indicating increased amplification redundancy and reduced effective library complexity.

Barcode analysis revealed broadly similar molecular complexity in sample1 and sample2. In contrast, sample3 exhibited approximately four to five-fold higher reads-per-barcode values, indicating substantially increased amplification redundancy (Figure 4). This suggests that a larger fraction of sequencing depth in sample3 originated from repeated observations of previously captured molecules rather than additional unique molecular events, resulting in reduced effective library complexity.

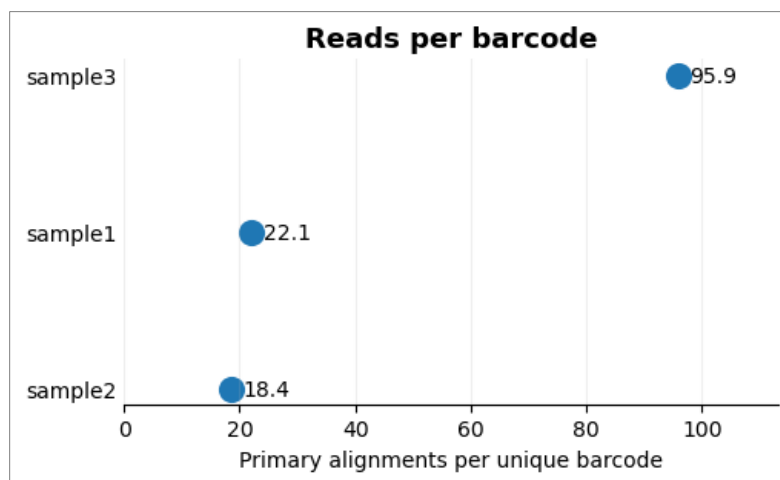


Figure 4. Barcode level amplification redundancy analysis. Amplification redundancy was quantified as the ratio of primary alignments to unique molecular barcodes (mb) within each sample (Primary Alignments / Unique Molecular Barcodes). This metric estimates the average number of sequencing alignments generated per originating molecule.

MOLECULE LEVEL ABUNDANCE

Deduplicated Count Abundance

Following quality control, enrichment analysis was performed to identify the genomic regions and genes preferentially captured by the targeted sequencing assay. Given the evidence of amplification redundancy observed during QC, enrichment was assessed using both raw alignment counts and deduplicated molecular counts. Comparing these measurements allowed the impact of amplification bias on apparent enrichment to be quantified before downstream biological interpretation.

Comparison of raw read counts and deduplicated molecular counts showed that enrichment estimates were substantially affected by amplification redundancy in sample3 (Figure 5). Many of the most highly enriched loci exhibited reads-per-molecule ratios exceeding 50–100, indicating that a large fraction of the observed sequencing depth originated from repeated sequencing of the same underlying molecules rather than additional unique molecular observations.

This finding is consistent with the earlier QC analysis, where sample3 also demonstrated elevated reads-per-barcode values and reduced effective molecular complexity. Despite these differences, the overall enrichment profile remained similar across samples, with many of the same dominant loci observed in all three libraries.

For this reason, sample3 was retained for downstream analysis. However, enrichment estimates were interpreted primarily using deduplicated molecular counts rather than raw sequencing depth. In contrast, samples1 and 2 retained substantially larger numbers of unique molecules across enriched loci and showed more moderate amplification redundancy overall.

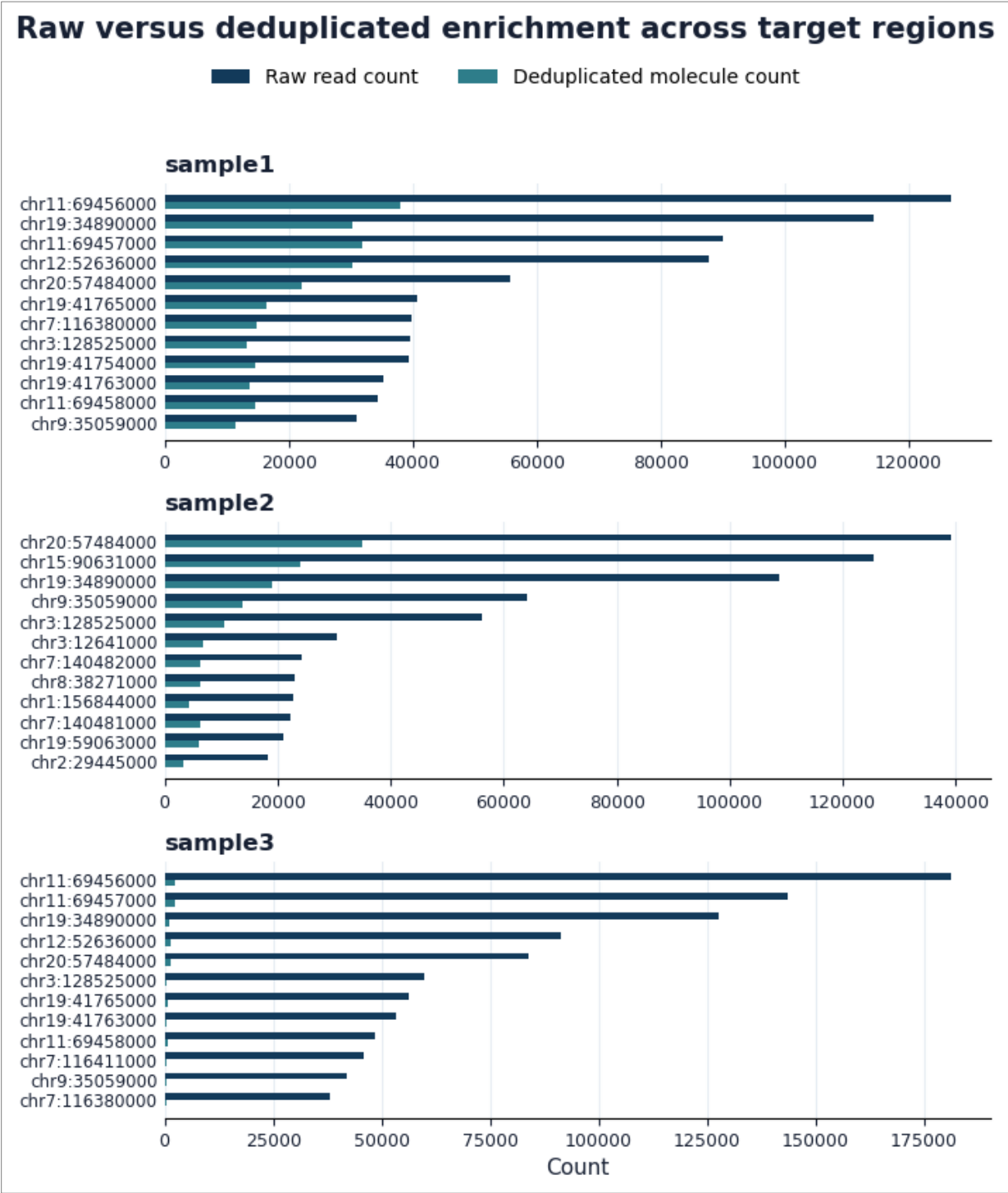


Figure 5. Raw versus deduplicated enrichment across target genomic regions. Raw alignment counts and deduplicated molecular counts are shown for the top enriched genomic bins in each sample. Deduplication was performed using the embedded molecular barcode (mb) annotations to collapse repeated observations of the same originating molecule.

Gene Enrichment and Annotation

To determine which genes were represented within the recurrently enriched genomic regions, the top barcode-deduplicated enriched bins were annotated against GENCODE v19 gene models using the hg19/GRCh37 reference coordinate system. Because no target BED file or assay annotation was provided with the assignment, enriched regions were first identified directly from the aligned BAM files and then intersected with gene-level annotations.

Annotation of the recurrently enriched regions identified a relatively focused set of protein-coding genes that were repeatedly detected across samples. The most strongly enriched loci included genes such as **GNAS**, **CCND1**, **IDH2**, **AXL**, **MET**, **EGFR**, **FGFR1**, **BRAF**, **HRAS**, **RAF1**, **MAP2K1**, and **RAB7A** (Figure 6).

Several genes, including **CCND1**, **AXL**, and **MET**, were represented by multiple adjacent enriched bins, suggesting that enrichment was distributed across broader genomic intervals rather than isolated single regions. Many of the identified genes participate in oncogenic signaling, cell-cycle regulation, and DNA damage checkpointing; pathways frequently altered in cancer. Although the intended target panel was not provided, the concentration of enrichment within a relatively small set of cancer associated genes is consistent with a targeted assay design rather than broad genome-wide sequencing.

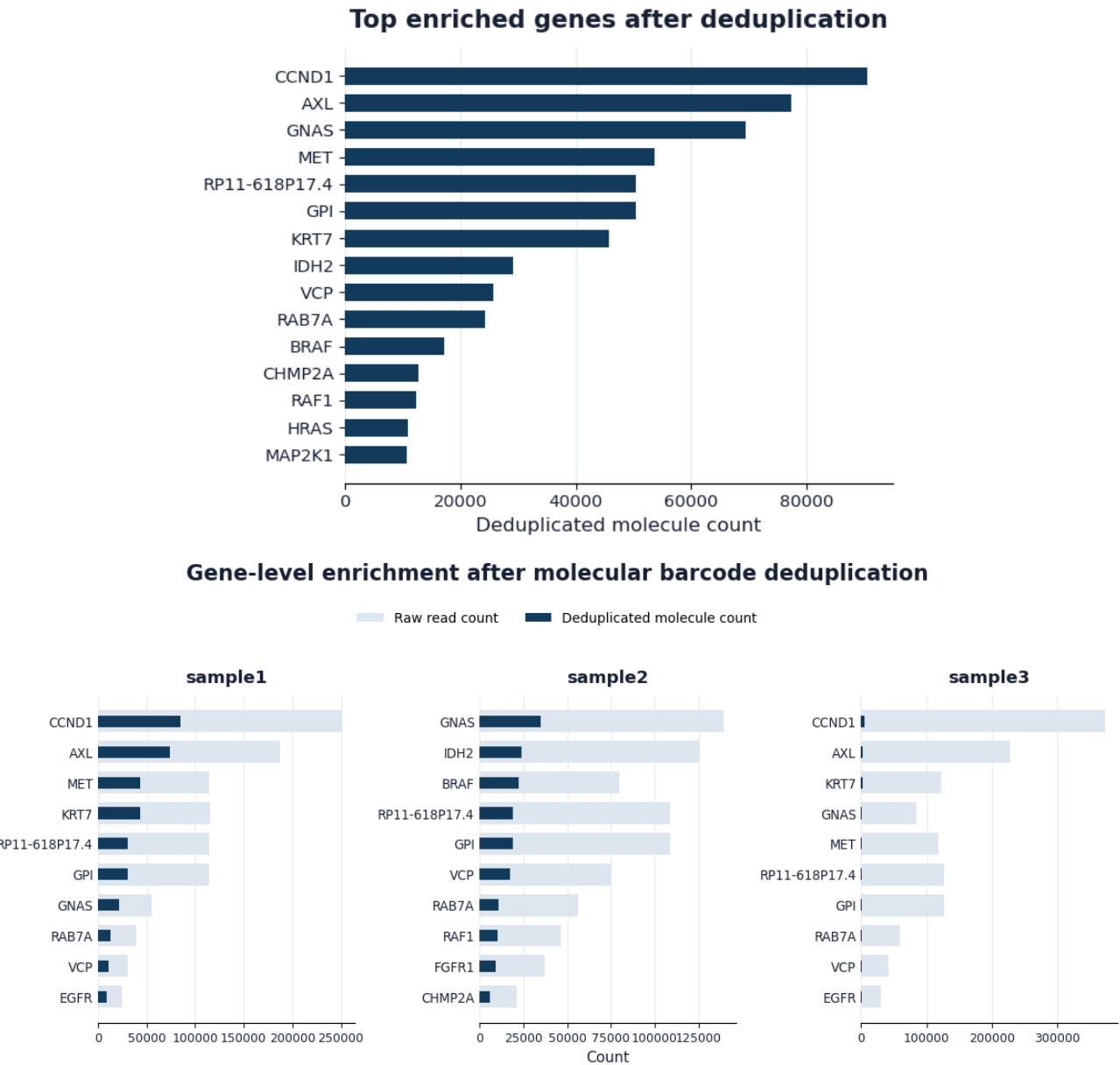


Figure 6. Gene-level enrichment following molecular barcode deduplication. The top panel shows the most strongly enriched genes identified after molecular barcode deduplication across all samples. Recurrently enriched loci included **CCND1, AXL, GNAS, MET, IDH2, BRAF, RAF1, HRAS, and MAP2K1**, many of which are involved in signaling pathways and cancer-associated biological processes. The lower panels compare raw read counts and deduplicated molecular counts for the top enriched genes within each sample. While molecular barcode deduplication reduced apparent enrichment across many loci, the overall enrichment profile remained consistent, with many of the same genes repeatedly detected across all three samples. These results suggest that the observed enrichment is reproducible across libraries and not driven solely by amplification-related artifacts.

CANDIDATE VARIANT ANALYSIS

Recurrent Variance within and across samples

Following quality control and enrichment analysis, sequence variation present within the enriched target regions was performed. Candidate variants were identified directly from the aligned BAM files and filtered using read depth and variant quality metrics prior to annotation.

Overall, a relatively small number of high-confidence candidate variants were detected across the dataset. Most variants were observed in only a single sample, suggesting that most sequence variation represents sample-specific events rather than recurrent features of the targeted assay. Sample2 contained the largest number of candidate variants, while samples1 and 3 exhibited fewer unique events overall.

The strongest recurrent signal was observed within FGFR1, where two adjacent nucleotide substitutions were detected in all three samples with substantial read-depth support (Figure 7). These variants represented the only sequence alterations shared across the entire dataset and therefore constitute the most reproducible variant-associated feature identified within the targeted regions.

Additional candidate variants were detected within several enriched genes, including AXL, MET, RAF1, EGFR, RAB7A, GPI, and CCND1 (Table 1). In contrast to FGFR1, these events were generally restricted to individual samples and did not demonstrate reproducibility across libraries.

Taken together, the variant landscape was characterized by a limited number of recurrent events superimposed upon a larger set of sample-specific candidates. The recurrence of the FGFR1 substitutions suggests that this locus represents a consistent feature of the dataset, although the absence of matched controls, assay design information, and orthogonal validation prevents definitive classification as biological or technical in origin.

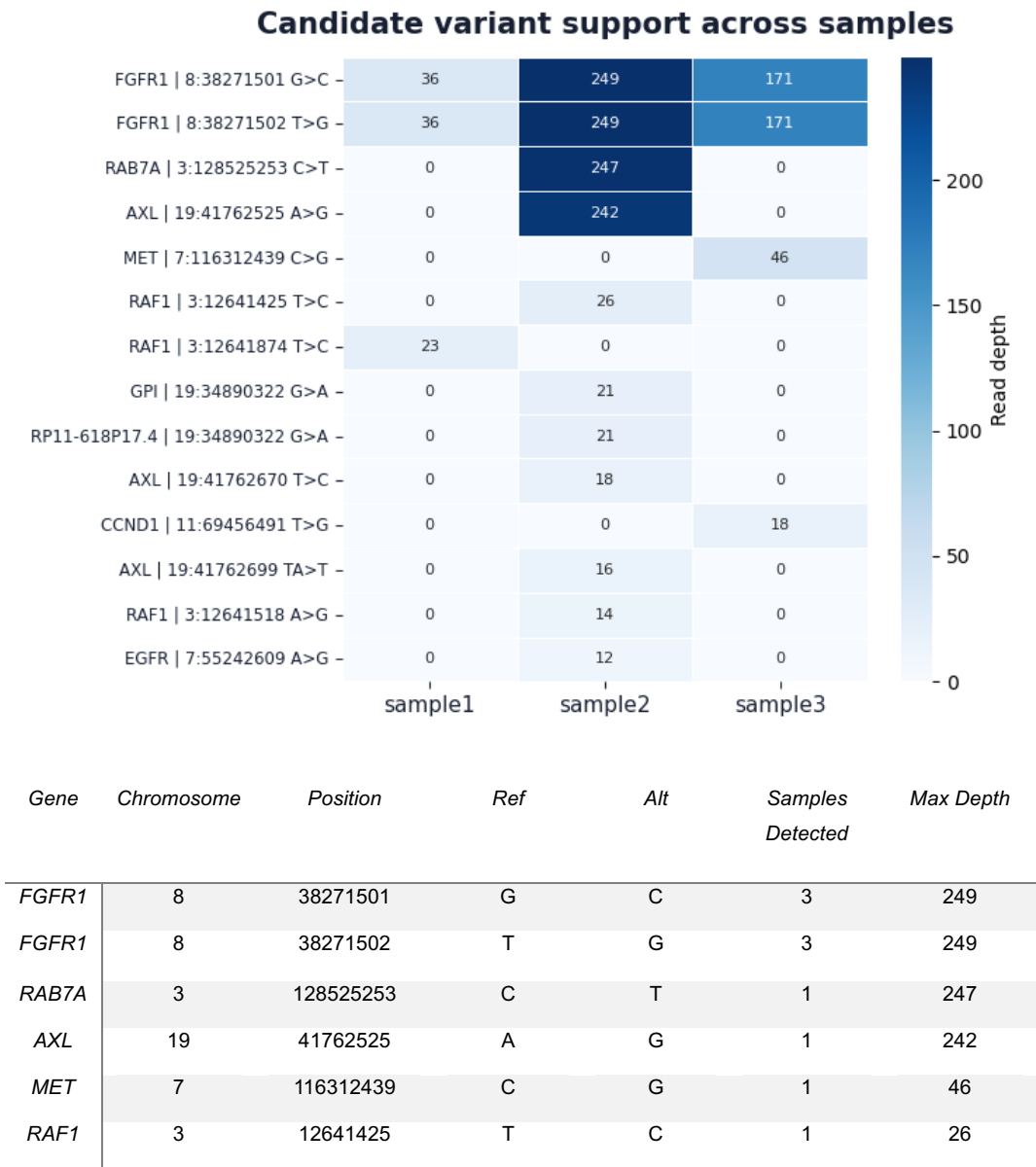


Figure 7. Recurrent candidate variants across samples. Heatmap showing read-depth support for filtered candidate variants identified within enriched target regions. Rows represent individual variants annotated to overlapping genes, while columns represent individual samples. Color intensity and embedded values indicate read depth supporting each variant call.

SOFT-CLIPPED READ CHARACTERIZATION

To further characterize the aligned reads, soft-clipped sequences present within the BAM files were analyzed. Soft clipping occurs when a portion of a sequencing read does not align to the reference genome and is therefore excluded from the primary alignment while the remaining sequence aligns successfully. CIGAR convention was used to compress soft-clipping analysis per read. Large soft-clipped segments can arise from a variety of sources, including alignment ambiguity, structural variation, repetitive sequence content, reference mismatches, or assay-specific sequencing artifacts.

Across all three samples, soft-clipped reads were readily detected but were concentrated within a limited number of recurrent genomic loci rather than being uniformly distributed throughout the genome. Instead, most large soft-clipped reads originated from a relatively small number of recurrent genomic loci. The most frequently observed annotated region mapped to CCND1, while several additional recurrent loci localized to poorly annotated or unplaced genomic regions, including multiple intervals on the GL000220.1 reference contig (Figure 8).

In absolute terms, sample2 contained the largest number of large soft-clipped reads, followed by sample3 and sample1 (Figure 8). However, when soft-clipped reads were normalized by barcode-deduplicated molecular abundance, sample3 exhibited substantially higher soft-clipping rates across several enriched genes, including HRAS, CCND1, and AXL. Notably, sample3 was independently identified during earlier QC and deduplication analyses as having the highest amplification redundancy and the lowest effective molecular complexity of the three samples.

Taken together, these findings indicate that soft clipping is concentrated within a limited number of recurrent genomic loci rather than being distributed uniformly throughout the dataset. Although sample2 contributed the largest absolute number of soft clipped reads, normalization by molecular abundance identified a disproportionate soft clipping burden in sample3. This observation is consistent with earlier analyses showing increased amplification redundancy and reduced effective molecular complexity in that sample.

Because many of the dominant soft clipped regions occur within poorly annotated loci and only aligned BAM files were available for analysis, the biological significance of these sequences remains uncertain. However, the recurrent nature of these events suggests that the observed soft clipping reflects systematic characteristics of the dataset rather than random alignment noise.

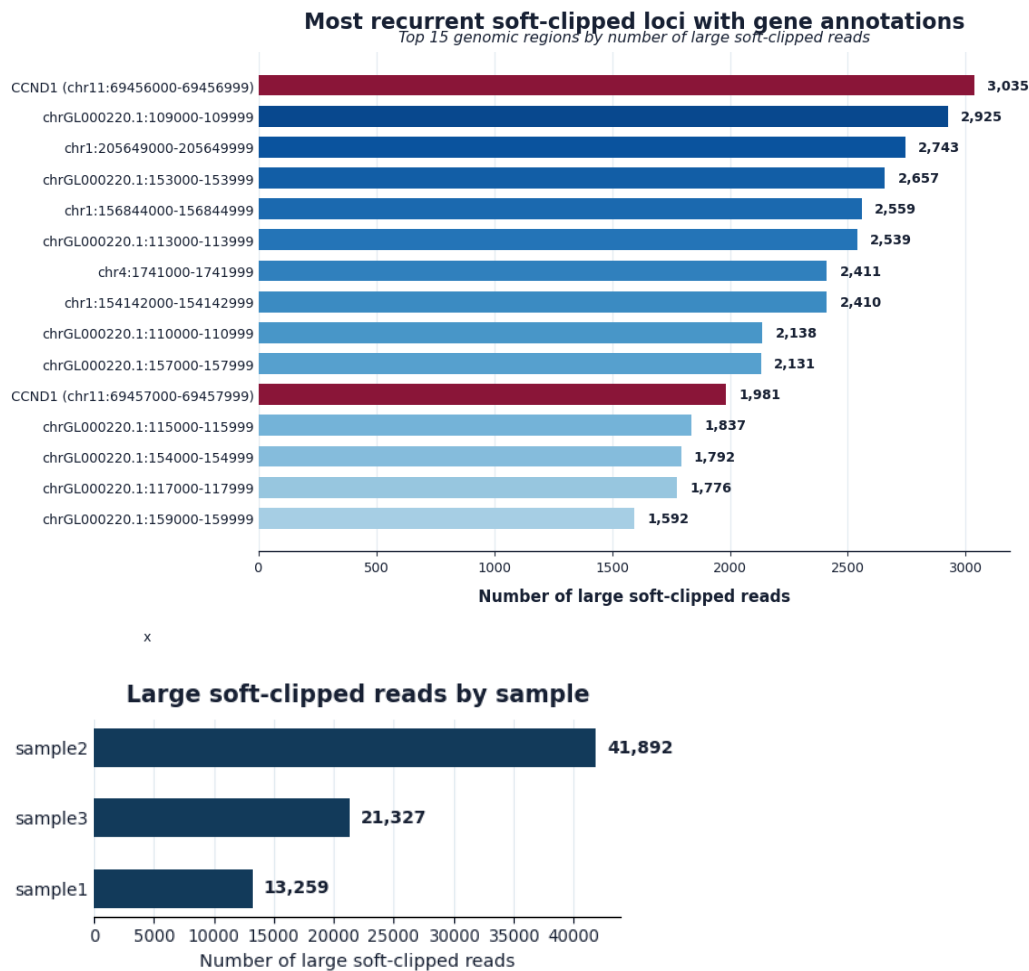


Figure 8. Recurrent large soft clipped loci identified across samples. (Top) Large soft clipped reads were aggregated across all samples and grouped by genomic location. The plot shows the fifteen most frequently observed loci associated with large soft clipped sequences. Recurrent soft clipping was concentrated within a limited number of genomic regions rather than being distributed uniformly throughout the dataset. (Bottom) The most frequently observed annotated locus mapped to CCND1, while several additional high frequency regions localized to the poorly annotated reference contig GL000220.1. These findings indicate that soft clipping is driven by a small number of recurrent loci and represents a systematic feature of the dataset rather than random alignment noise.

DISCUSSION

The dataset was technically consistent across all three samples. Alignment quality was strong, read-length distributions were reproducible, and enrichment was concentrated within a relatively small number of genomic regions. Despite differences in sequencing depth and barcode complexity, the same major targets were recovered across samples, supporting a focused enrichment strategy rather than broad, nonspecific coverage.

The largest source of variation was molecular complexity. Barcode deduplication had modest effects in sample1 and sample2 but substantially reduced abundance estimates in sample3. Elevated reads-per-barcode values, fewer unique molecules, and large differences between raw and deduplicated counts all point to increased amplification redundancy within that library. These results highlight how heavily quantitative interpretation can depend on molecule-level support.

Enrichment was concentrated within a relatively small number of recurrent loci associated with cancer. **CCND1**, **AXL**, **MET**, **EGFR**, **FGFR1**, **BRAF**, **HRAS**, and **MAP2K1** remained among the dominant targets before and after deduplication, indicating that the overall enrichment profile was not driven solely by amplification artifacts. Although the target panel was not provided, the concentration of signal within this group of genes is consistent with a clinically focused targeted assay.

Sequence variation within enriched regions was limited. Most candidate variants were restricted to individual samples, while two adjacent **FGFR1** substitutions were detected across all three libraries with consistent read-depth support. **FGFR1** was the only locus showing reproducible variant-associated signal across the dataset. Whether these events represent biological variation or assay-specific characteristics cannot be determined from the available data.

Soft-clipped reads were also concentrated within a limited number of recurrent loci. Sample2 contained the largest absolute number of soft-clipped reads, but normalization by deduplicated molecular abundance again highlighted sample3 as distinct from the other libraries, particularly within **HRAS**, **CCND1**, and **AXL**. The combination of elevated

soft clipping, increased amplification redundancy, and reduced molecular complexity suggests that sample3 contains sequence features not observed to the same extent in the other samples. While the current data do not support a structural rearrangement or gene fusion call, the pattern may warrant additional investigation.

Only aligned BAM files were available for analysis. Raw FASTQ files, assay design information, target BED files, and matched controls were not available. As a result, several observations could be characterized but not fully resolved. Even with those limitations, the dataset provides a clear picture of target enrichment, molecular complexity, sequence variation, and alignment behavior while demonstrating the value of molecular barcode information for downstream interpretation.

CONCLUSION

The analyses presented here indicate that the dataset is technically sound and captures a focused set of recurrent genomic targets with strong consistency across samples. While the overall enrichment landscape was highly reproducible, incorporation of molecular barcode information revealed important differences that were not apparent from sequencing depth alone.

The most notable observation was the distinct behavior of sample3, which consistently exhibited reduced molecular complexity, elevated amplification redundancy, and increased normalized soft-clipping rates. Together, these findings highlight the importance of evaluating molecule-level support when interpreting targeted sequencing data and demonstrate how reliance on raw read counts alone can lead to misleading conclusions.

Although several candidate variants and recurrent sequence features were identified, the available data were insufficient to determine their biological significance with confidence. Despite these limitations, the major technical features of the dataset were readily apparent. More broadly, this assessment demonstrates how molecular barcodes can be leveraged to separate technical artifacts from true biological signal and prioritize features for downstream investigation.